

SANDALI RUWANYA

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PROFILE

Data Science undergraduate at SLIIT with hands-on experience across the full data stack, from real-time streaming pipelines and machine learning to enterprise BI reporting and full-stack web development. Skilled in building ETL pipelines, OLAP cubes, Power BI dashboards, Kafka-based streaming systems, and production-grade REST APIs with advanced security features.

EDUCATION

B.Sc.(Hons) in Information Technology — Data Science — SLIIT

2023 – Present

Advanced Level Studies — St. Paul's Girls' School, Milagiriya

2020 – 2022

TECHNICAL SKILLS

Data & BI:	SQL Server · SSIS (ETL) · SSAS (Cubes · MDX) · Power BI · Dimensional Modelling · Star Schema · OLAP
Data Science:	Python (pandas, numpy, scikit-learn) · R (dplyr, ggplot2) · Statistical Testing · Regression · EDA
Data Engineering:	Apache Kafka · Docker · PostgreSQL · SQLAlchemy · cron Automation
Backend:	Java · Spring Boot · REST APIs · Spring Security · JWT · Google OAuth 2.0 · Two-Factor Auth (OTP) · SSE
Frontend:	React.js · Tailwind CSS · Recharts · Axios · Node.js · Express.js
Databases:	Microsoft SQL Server · PostgreSQL · MongoDB Atlas · MySQL
Tools:	Git · GitHub · GitHub Actions (CI/CD) · Postman · Jupyter · VS Code · Figma · Microsoft Excel

PROJECTS

Real-Time IoT Streaming Pipeline

2026

Apache Kafka · Python · PostgreSQL · Docker · Grafana

- Built an end-to-end real-time data pipeline simulating IoT sensors, streaming events through a 3-partition Kafka topic into PostgreSQL with sub-5-second latency
- Implemented producer/consumer architecture with data validation, fault-tolerant consumer groups, and end-to-end latency tracking
- Built a live Grafana dashboard with 5-second auto-refresh visualizing sensor data across multiple locations

Cryptocurrency Analytics ETL Pipeline

2026

Python · PostgreSQL · pandas · SQLAlchemy · CoinGecko API · Power BI · cron

- Built an automated end-to-end ETL pipeline ingesting live market data for 5 major cryptocurrencies (BTC, ETH, BNB, SOL, ADA) from the CoinGecko API on an hourly cron schedule
- Designed a star schema data warehouse in PostgreSQL with SQLAlchemy-managed ingestion, separating raw, staging, and analytics-ready layers
- Built a Power BI dashboard surfacing price trends and market movement, enabling scheduled batch analytics as a complement to real-time streaming approaches

Social Media Campaign Engagement Predictor

2026

Python · pandas · scikit-learn · Random Forest

- Engineered marketing KPIs (CTR, CPC, CPM, ROAS) from 300,000 raw ad campaign records across 4 platforms
- Trained a Random Forest classifier achieving 0.73 ROC-AUC to predict campaign engagement before launch, identifying CPC as the strongest predictor (43% feature importance)
- Delivered platform-level business recommendations, including identifying a critically underperforming channel costing 5x in lost revenue per campaign

Smart Campus Operations Hub — Auth & Notifications Module

2026

Java Spring Boot · React.js · MongoDB · Spring Security · JWT · Google OAuth 2.0 · SSE · GitHub Actions

- Built complete authentication system: Google OAuth 2.0, Two-Factor Authentication (OTP via email), and JWT issuance — genuine MFA exceeding the OAuth-only requirement
- Engineered real-time notification system using Server-Sent Events (SSE) with parallel email delivery, full Notifications REST API, and Audit Log tracking all admin actions
- Developed Admin Dashboard with Recharts analytics and configured GitHub Actions CI/CD pipeline

CERTIFICATIONS

MongoDB Data Modeling Path · MongoDB Basics for Students · MongoDB Indexes — *MongoDB University, 2026*

SOFT SKILLS

Team Collaboration · Problem Solving · Communication · Time Management · Attention to Detail